

SHRUTI SINHA

Data & AI Engineer | Power BI | SQL | Python | Azure | Generative AI

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Profile

Senior Data & AI Engineer with 8+ years of experience in data engineering, database development, analytics, and cloud-based data solutions. Strong hands-on experience designing scalable ETL/ELT pipelines and data platforms using Python, SQL, Microsoft Azure, Databricks, and PySpark. Experienced in Generative AI, LLM applications, RAG, prompt engineering, AI agents, MCP, embeddings, and vector-based retrieval, with a focus on integrating AI capabilities with enterprise data platforms and building scalable, production-ready data and AI solutions.

Skills

Programming & Data Engineering

Python, SQL, PySpark, Apache Spark, Pandas, NumPy, Scikit learn, ETL/ELT, Data Pipelines, Data Modeling, Data Warehousing, TensorFlow, PyTorch

Generative AI & LLM

Generative AI, LLMs, Prompt Engineering, RAG, Agentic AI, MCP, LangChain, LangGraph, CrewAI, Embeddings, Vector Databases

Cloud & DevOps

Microsoft Azure, Azure Data Factory (ADF), ADLS Docker, Kubernetes, Azure DevOps, CI/CD, AWS

Backend & APIs

FastAPI, Flask, REST APIs, API Integration, JSON, Authentication & Authorization

Education

MS/M.Sc., Hochschule heilbronn, Germany, Software Engineering (1.3) 2024 – 2026

B.Tech, West Bengal University of Technology (WBUT)

Work Experience

SENIOR DBA, Encora 03/2022 – 12/2024

Project: Sales Data & Generative AI Platform | Azure & GenAI | Team Size: 3

Built a scalable Azure data platform for sales analytics and AI use cases, developing ETL/ELT pipelines using ADF, ADLS, Databricks, PySpark, Python, and SQL, and integrating GenAI, RAG, LLMs, and agentic AI workflows with enterprise applications.

Project: Customer Support Chatbot | RAG, LLM | Team Size: individual

Built a RAG pipeline for a customer chatbot involving document processing, chunking, embeddings, vector database retrieval, semantic search, and LLM-based response generation.

- Implemented data quality and validation checks to identify anomalies, inconsistencies, and pipeline issues, improving data accuracy and process reliability. Followed CI/CD and DevOps practices using Git/GitHub for testing.
- Investigated data and pipeline issues using logs, monitoring, and validation results, performed root cause analysis and implemented corrective and preventive solutions. Containerized AI applications using Docker with Kubernetes.
- Collaborated with cross-functional teams and developed LangChain/LangGraph workflows to deliver maintainable AI solutions aligned with business requirements.

- Built and evaluated GenAI workflows using Azure AI Foundry, MLflow, Databricks AgentBricks, Genie, and PyTorch, supporting LLM experimentation, model tracking, agent development, and AI solution deployment.

Senior Analyst, Ocwen Financial Corporation

01/2020 – 02/2022

Project: Business Process Automation | Python & VBA | Team Size: individual
Automated business, reconciliation, validation, and reporting workflows using Python, SQL, and VBA, improving operational efficiency by 70% and significantly reducing manual effort.

Project: Business Process Automation | SQL & VBA

Built interactive Power BI and Tableau dashboards for KPI monitoring, trend analysis, and business reporting, enabling stakeholders to derive actionable insights from operational data

- Developed Python and SQL data processing, reconciliation, and data quality workflows.
- Developed data-processing and reporting workflows to improve data accuracy and consistency.
- Investigated production data issues using data validation and root-cause analysis.

Consultant, JPMorgan Chase Bank

05/2019 – 11/2019

- Performed SQL and Cognos-based validation and analysis of customer data, improving data accuracy, consistency, and compliance with business requirements.
- Collaborated with business analysts and technical teams to translate business requirements into data solutions and resolve complex data-related challenges.

Business Analyst, Breakwalls Technologies

03/2018 – 04/2019

- Worked on Supply Chain Management solutions, consolidating data from multiple SQL Server databases to support business reporting and analysis.
- Developed and optimized SQL procedures, triggers, queries, and data models for business intelligence, reporting, and downstream analytics.
- Resolved production database issues and optimized SQL scripts to maintain reliable and efficient data processing.

Information Executive, Tracxn

06/2016 – 02/2018

- Conducted sector-focused research across Healthcare, Banking, and Retail Tech, analyzing startup capitalization structures, stakeholder information, investments, and funding rounds.
- Trained and mentored newly joined research executives on industry research methodologies, data analysis across Finance and Healthcare.

Thesis Projects

Human Pose Estimation | Team Size: individual

Technologies: Python, OpenCV, MediaPipe, Scikit-learn, Computer Vision, Machine Learning

- Developed an end-to-end computer vision and machine learning pipeline using Python, OpenCV, MediaPipe, and Scikit-learn for workout pose detection and classification.
- Improved classification performance through controlled augmentation and model evaluation, achieving 94.44% test accuracy compared with 77.78% on the original dataset.

Certificate

- Microsoft Certified: Azure Data Engineer Associate(DP-203)
- Google Project Management Certification
- Develop Generative AI Applications from IBM